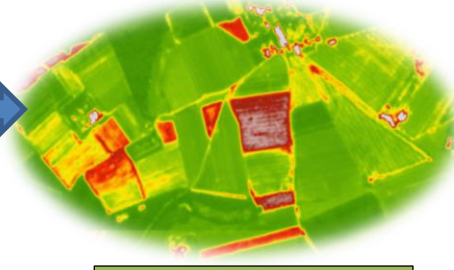


Stage:1 Development of Relation between NDVI & Tomato Yield



Satellite Image of Tomato farms

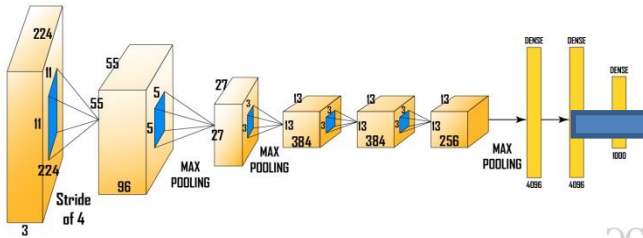


NDVI Value of Tomato

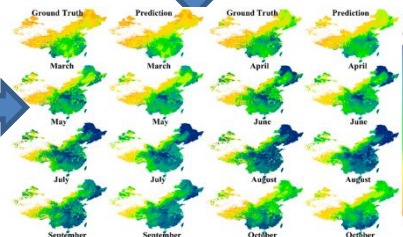


Development of analytical relation
NDVI and Tomato

Stage:2 Development of MRCNN model



Development of MRCNN model

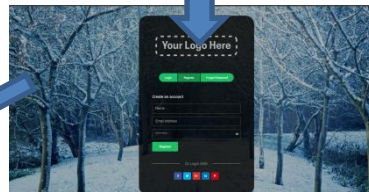


Prediction of NDVI for the Tomato Yields



Prediction of Tomato Yields

Stage:3 Effective e Tomato Supply Chain management



AI-TOPSCS: (e-market for Tomato products)



Tomato Farmers



Tomato Sellers



Tomato Cold storage



Value added Products

AI-TOPSCS: AI-Based Tomato operation, production, and supply chain system.

Abstract:

In the future, the Tomato supply chain (TSC) will encompass several crucial stages requiring planning at least two months ahead. An accurate tomato yield forecast well in advance will be essential for making timely and informed decisions regarding resource allocation, transportation logistics, and cold storage contracts in the TSC. Any overestimation or underestimation of tomato yield could lead to significant inefficiencies. Recognizing its importance, we will propose a method based on machine learning (ML) techniques that leverages satellite-based vegetation data. This method, known as the Satellite Image-Based Method for Tomato Production Forecast (SIBM-TPF), will delve into Normalized Difference Vegetation Index (NDVI) data collected at various phenological stages to capture spatial and temporal dependencies in tomato yield forecasting.

In the initial phase of SIBM-TPF, a clustering process will be employed, utilizing georeferenced NDVI data from all phenological stages as input, resulting in the identification of several clusters. Subsequently, in the second step, two validation functions will be applied to determine the optimal clustering approach. Finally, in the third step, the predictor variables from the best clustering will be incorporated into an artificial neural network (ANN) to predict tomato yield. SIBM-TPF will be successfully applied to forecast the table of tomato yields in a selected region of the country. We expect the results to be 95% accurate for forecasting tomato yield at least two months in advance. These results will be incorporated into our e-market service web platform for effective supply chain management of tomatoes. The implementation of SIBM-TPF will offer stakeholders the opportunity to make more informed decisions, enhancing the coordination of TSC stages while reducing costs and minimizing tomato losses.

Description of Idea.

In this study, Sentinel-2 satellite data, along with Vegetation Indices (VIs), will be employed to predict future tomato yields. Data will undergo pre-processing and statistical analysis to establish forthcoming yield relationships. A Recurrent Convolutional Neural Network (RCNN) model will be trained for yield prediction, and the data will be integrated into future supply chain management for efficient tomato distribution..

The working principle of the system is shown in figure 1

Data Acquisition:

In this study, remote sensing data will consist of multispectral images with a resolution of $10 \times 10 \text{ m}^2$, which will be obtained from the multispectral imager (MSI) sensor on ESA's Sentinel-2 satellite platforms. We will acquire atmospherically corrected imagery (level 2A), providing reflectance values at the bottom of the atmosphere, in cartographic geometry (UTM/WGS84 projection) from the ESA portal (<https://scihub.copernicus.eu/>). Satellite imagery data will be collected at 5-day intervals, and for each date, we will download a set of three (3) satellite images to cover the entire study area.

Data Preprocessing:

The initial preprocessing step will involve resampling the images to a consistent 10-meter resolution. This will be necessary because Sentinel-2's spectral bands operate at varying spatial resolutions: 10 meters, 20 meters, and 60 meters. Additionally, on days with heavy cloud cover obscuring our fields of interest, corresponding values will be removed from the dataset to maintain data integrity.

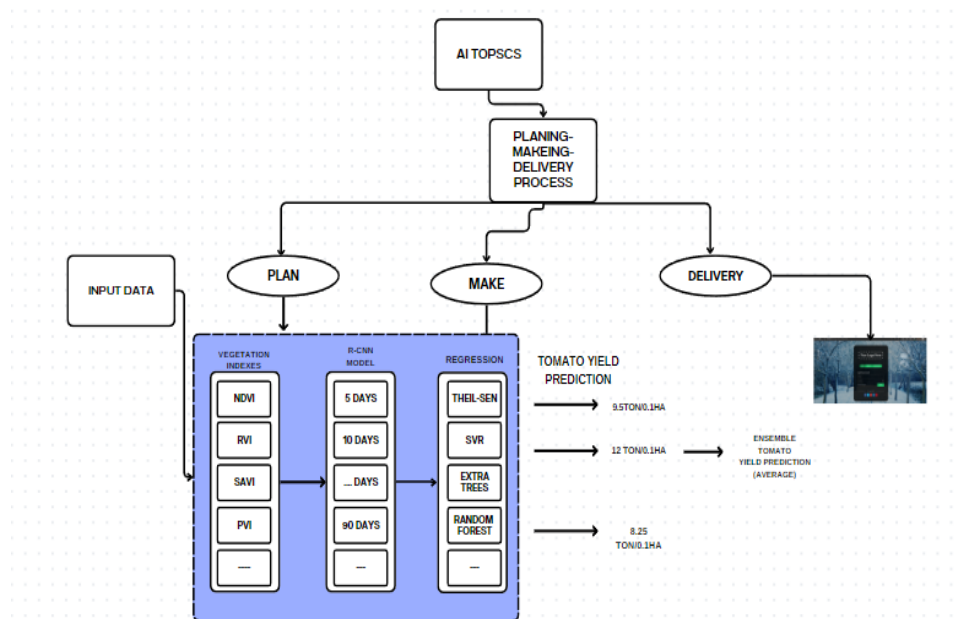


Figure 1 AI based Tomato operation,production and supply chain system.(AI-TOPSCS)

Vegetation Indices (VIs):

To assess the relationship between satellite data and crop yield, we will calculate five Vegetation Indices (VIs) for each date: the normalized difference vegetation index (NDVI), weighted difference vegetation index (WDVI), soil-adjusted vegetation index (SAVI), ratio vegetation index (RVI), and perpendicular vegetation index (PVI). This will result in a VI raster dataset.

Subsequently, we will gather actual yield data and conduct statistical analyses to evaluate and establish relationships between these yield figures and the calculated VIs.

RCNN Model Training:

The next step involved training and testing a Recurrent Convolutional Neural Network (RCNN) model to predict tomato yields using the acquired data.

The proposed framework for automatic and simultaneous individual tomato crop delineation and tree height, NDVI assessment will be based on the Mask R-CNN model. This model will be capable of classifying an object at the pixel level and delineating its shape, providing three outputs for each object: a class label, a bounding box, and a mask. In this study, the class label will represent the predictive crop height, NDVI and rate of production.

Supply Chain Management:

Finally, the predicted yield data were integrated into planning, storage, and delivery models to enhance the effective management of the tomato supply chain. To connect the planning predicting, storage and delivery process the web based e-platform is created.

The analysis of historical demand data and the output of AI-predicted models will be used to forecast future customer demand, optimizing resource planning, planning marketing actions and storage.

The analysis of historical data will be used to identify relationships between costs and product characteristics in order to predict future industrial costs of new build-to-order products and generate reliable quotations